

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Rear Crankshaft Seal Replacer Kit, Part Number 3164660, for unitized rear crankshaft seals **only**
- Rear Crankshaft Seal Installation Kit, Part Number 5395259, for rear crankshaft lip seals **only**
- Rear Crankshaft Seal Removal Kit, Part Number 5395330, for rear crankshaft lip seals **only**
- Rear crankshaft seal carrier centering guide, Part Number 5395409, for front gear train engines **only**
- P-80 lubricant, Part Number 3824878, or mild soapy water

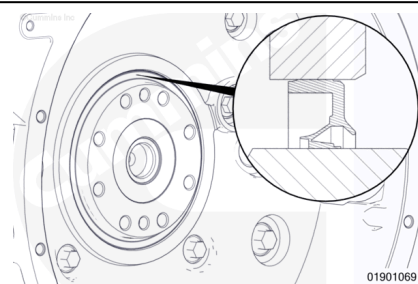
Additional Service Items

- Safety solvent
- Anti-seize compound or a suitable grease

General Information

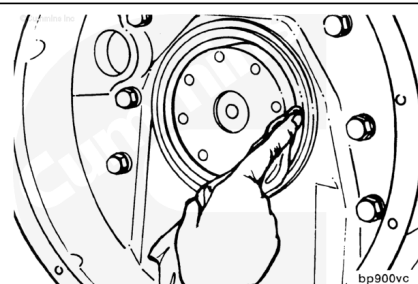
Lip Seal

Engines covered by this procedure use a lip style rear crankshaft seal (shown). The rotating portion of the sealing occurs at the contact surface between the lip of the seal and the crankshaft. The seal is installed into the flywheel housing/seal carrier bore using required service tools, Rear Crankshaft Seal Installation Kit, Part Number 5395259, and Rear Crankshaft Seal Removal Kit, Part Number 5395330.



Some rust prevention oils applied at the factory have fluorescent tracer dyes added to them and can provide false test results when checking for external oil leaks using a black light.

Clean the area around the rear seal with solvent.



Check the area with a black light to insure there is **not** already fluorescent dye present before adding tracer dye to the lubricating oil.

Unitized Seal

Some engines used to use a unitized seal, which has a metal inner diameter that allows the inner and outer diameter of the seal to press-fit onto the crankshaft and the flywheel housing respectively, requiring service tool, Part Number 3164660, to remove.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Use a container that can hold at least 26 liters [27 U.S. qt] of lubricating oil.

- Disconnect the batteries. See equipment manufacturer service information.
- If equipped with a wet flywheel housing, drain the oil from the flywheel housing by removing the plug in the bottom of the flywheel housing.
- Remove the transmission and all related components, if equipped. See equipment manufacturer service information.
- Remove the flywheel, if installed. Reference Procedure 016-005 in Section 16.
- Remove the flexplate, if installed. Reference Procedure 016-004 in Section 16.

Remove

Lip Seal

CAUTION

This procedure must not be used on engines with a unitized seal.

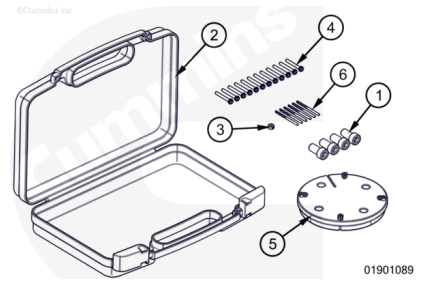


Table 1, Rear Crankshaft Seal Removal Kit, Part Number 5395330

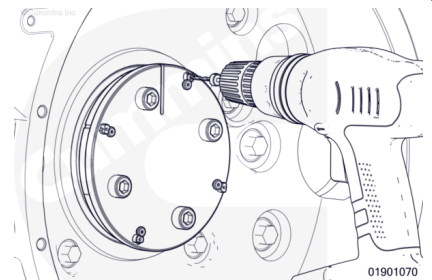
Item Number	Part Number	Description	Quantity
1	3164174	Screw, Socket Head Cap	4
2	3165143	Case, Storage	1
3	5395324	Collar, Stop	1
4	5395325	Screw, Self Tapping Metal	12
5	5395329	Remover, Crankshaft Seal	1
6	5395410	Bit, Drill	6

Use the groove on the face of the rear crankshaft seal remover to set the depth on the drill bit stop collar.

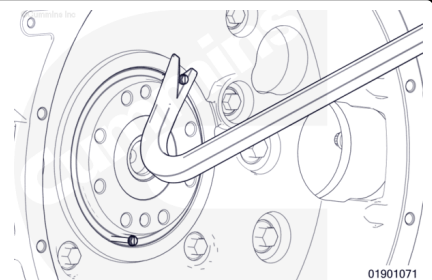
Install the crankshaft seal remover toward the end of the crankshaft with socket head capscrews.

Use the stop collar and drill bushings to drill two holes 180 degrees apart.

Remove the crankshaft seal remover from the crankshaft.



Use a driver to install the self tapping metal screw into the drilled holes. Use a forked prybar to pull the rear crankshaft seal out.



Unitized Seal

For engines with unitized seals, use service tool Part Number 3164660.

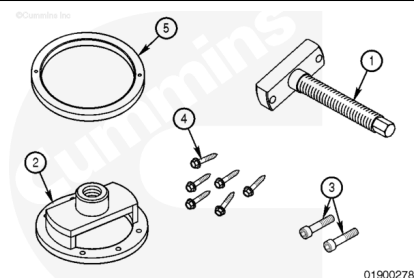
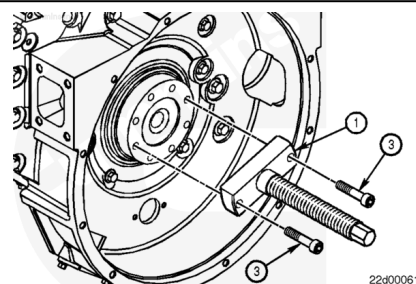


Table 2. Rear Crankshaft Seal Replacer Kit, Part Number 3164660

Item Number	Part Number	Description	Quantity
1	3164666	Replacer screw assembly	1
2	3164664	Crankshaft seal replacer	1
3	3164174	Socket head capscrew, M12 x 1.25x 25 mm	2
4	3164217	Sheet metal screw, Number 10x25.4 mm [1 in] long	25 (7 shown)
5	-	Disposable plastic driver (purchased with rear crankshaft seal kit)	1

With the rear crankshaft seal installed onto the crankshaft flange as described earlier in this procedure, mount the replacer screw assembly (1) onto the rear of the crankshaft. Install the two M12 x 1.25 x 60 mm socket head capscrews (3).

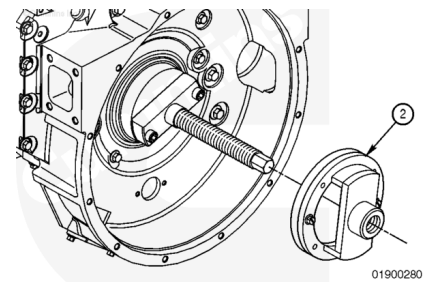


⚠ CAUTION ⚠

Do not use an impact wrench or air tools. Doing so can damage the tool.

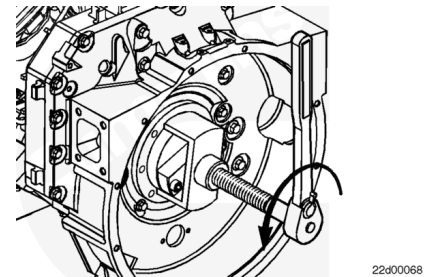
Lubricate the replacer screw with anti-seize compound or a suitable grease.

Hold the replacer screw and install the crankshaft seal replacer (2) onto the replacer screw assembly. Advance the crankshaft seal replacer toward the seal by rotating it **clockwise** until the attached disposable driver is positioned against the rear crankshaft seal. Then insert small screws into the crankshaft seal replacer tool.



⚠ CAUTION ⚠

Do not overtighten the replacer screw assembly after the crankshaft seal replacer contacts the flywheel housing. Doing so can damage the tool.



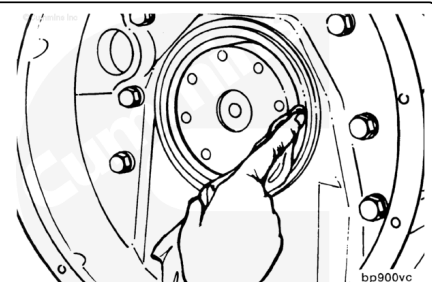
While holding the crankshaft seal replacer, rotate the replacer screw **counterclockwise** until the disposable driver attached to the crankshaft seal replacer makes contact with the flywheel housing.

Remove the service tools.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Use only with protective clothing, goggles/shield, and gloves to reduce the possibility of personal injury.

⚠ CAUTION ⚠

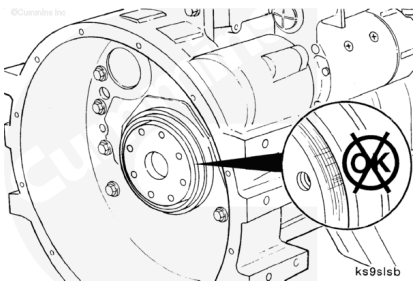
The seal lip and the sealing surface on the crankshaft must be free from all oil residue to prevent seal leaks.

Clean the rear crankshaft sealing surface and bore with solvent.

Dry the surface with compressed air.

If the crankshaft has excessive wear, a service wear sleeve is available.

Inspect the crankshaft seal contact area for a wear groove. If the groove is deeper than 0.25 mm [0.010 in], a wear sleeve and oversize seal **must** be used. Reference Procedure 001-067 in Section 1.

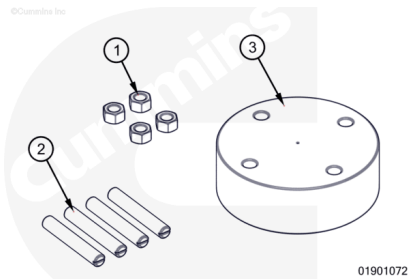


Install

For engines using a lip seal rear crankshaft seal, the use of Rear Crankshaft Seal Installation Kit, Part Number 5395259, is required.

Table 3, Rear Crankshaft Seal Installation Kit, Part Number 5395259

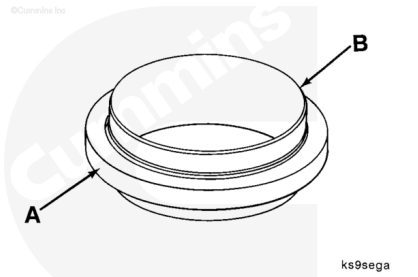
Item Number	Part Number	Description	Quantity
1	3163741	Nut, Regular Hexagon	4
2	5395257	Pin, Guide	4
3	5395258	Installer, Crankshaft Seal	1



⚠ CAUTION ⚠

The seal lip/bore and the sealing surface on the crankshaft must be free from all oil residue to prevent seal leaks.

To aid in installation, apply a mild soap (recommended 50/50 water to soap volumetric ratio) or P-80, Part Number 3824878, to the outside diameter of the seal case (A)

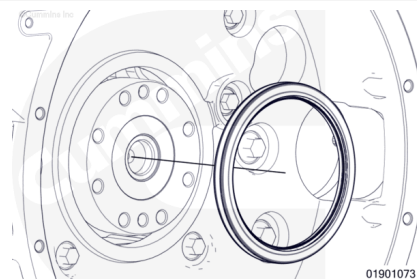


A seal pilot (B) is provided with the new seal. The seal **must** be left on the seal pilot while installing the seal onto the nose of the crankshaft so the lips of the seal do **not** get damaged during installation.

Place the new rear crankshaft seal, with tapered end of the seal pilot, over the crankshaft nose and slide it by hand toward the flywheel housing.

Verify the seal is positioned square with the crankshaft.

Remove the seal pilot.



Note : For rear gear train engines, the rear crankshaft seals for the wet flywheel housing applications do **not** come with a disposable seal driver. Service tool, Part Number 3824078, must be used to install the rear crankshaft seal for the wet flywheel housing. The same service tool is used for installing the rear crankshaft seal and wear sleeve assembly. Reference Procedure 001-067 in Section 1.

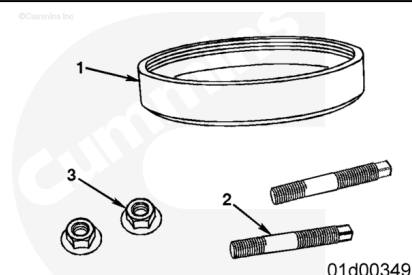
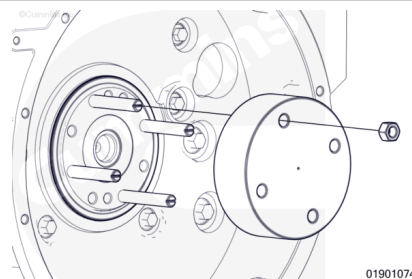


Table 4: Rear Crankshaft Seal Installation Kit, Part Number 3824078

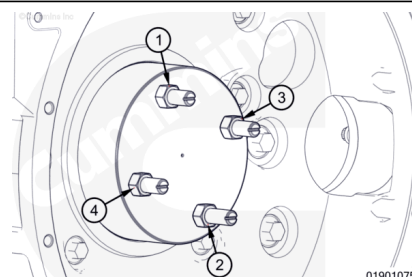
Item Number	Part Number	Description
1	3163734	Plate
2	3163628	Stud
3	3163741	Hex Nut

Install the rear crankshaft seal using Rear Crankshaft Seal Installation Kit, Part Number 5395259.

Install guide pins toward the end of the crankshaft with the slotted side facing away from the engine.



Note : On C Series and L Series engines **only**, if the seal carrier is removed for any repair purposes, use centering guide service tool, Part Number 5395409, to align the seal carrier to the cylinder block with reference to the centerline of the crankshaft. Reference Procedure 001-104 in Section 1.



Slide the crankshaft seal installer over the guide pins until it is touching the rear crankshaft seal.

Install the hex nuts onto the guide pins until the hex nuts bottom out on the crankshaft seal installer.

Use a deep well socket and tighten the hex nuts in a crisscross pattern until the crankshaft seal installer bottoms out on the crankshaft. Tighten all four nuts to the specification below to verify the crankshaft seal installer bottoms out with the crankshaft.

Torque Value: 10 to 12 n•m [89 to 106 in-lb]

The rear crankshaft seal installs deeper using the Crankshaft Seal Installation Kit compared to disposable service kits.

Finishing Steps

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the flywheel, if removed. Reference Procedure 016-005 in Section 16.
- Install the flexplate, if removed. Reference Procedure 016-004 in Section 16.
- Install the transmission and all related components, if equipped. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Start and operate the engine. Check for leaks.

